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Airfoil thickness profile for minimizing tip leakage flow

Abstract

A turbine engine assembly includes at least one rotor that has a plurality of blades, each of the blades includes an airfoil that has a pressure side and a suction side that each extend between a leading edge, a trailing edge, a tip and a base. The airfoil has a thickness between the pressure side and the suction side perpendicular to a camber line that varies between the leading edge and the trailing edge. The thickness includes a suction side thickness between the camber line and the suction side and a pressure side thickness between the camber line and the pressure side. A maximum ratio of the pressure side thickness to the suction side thickness is between 3 and 7.

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Background/Summary

TECHNICAL FIELD
(1) The present disclosure relates generally to a rotating airfoil utilized in a turbine engine. More

particularly, this disclosure relates to an airfoil for reducing leakage flow between a tip of the rotating airfoil and a static structure.

BACKGROUND

(2) A gas turbine engine typically includes a fan section, a compressor section, a combustor section and a turbine section. The compressor and turbine sections include rotating blades within a case structure. A clearance between tips of the rotating blades and the case structure is a source of leakage flow that decreases engine efficiency. Leakage flow is driven by a pressure differential across the tip. The clearance between tips of the rotating blades can be minimized but some minimal clearance is required to accommodate relative movement during operation. Moreover, the rotating blades are sized with a minimum thickness for durability and to provide desired airflow characteristics. Blade efficiency is improved by reducing leakage flow.

SUMMARY

(3) A turbine engine assembly according to an exemplary embodiment of this disclosure, among other possible things includes at least one rotor that has a plurality of blades, each of the blades includes an airfoil that has a pressure side and a suction side that each extend between a leading edge, a trailing edge, a tip and a base. The airfoil has a thickness between the pressure side and the suction side perpendicular to a camber line that varies between the leading edge and the trailing edge. The thickness includes a suction side thickness between the camber line and the suction side and a pressure side thickness between the camber line and the pressure side. A maximum ratio of the pressure side thickness to the suction side thickness is between 3 and 7.

(4) A blade for compressor section of a turbine engine assembly according to another exemplary embodiment of this disclosure, among other possible things includes an airfoil that has a pressure side and a suction side that each extend between a leading edge, a trailing edge, a tip and a base. The airfoil has a thickness between the pressure side and the suction side perpendicular to a camber line that varies between the leading edge and the trailing edge. The thickness includes a suction side thickness between the camber line and the suction side and a pressure side thickness between the camber line and the pressure side. A maximum ratio of the pressure side thickness to the suction side thickness is between 3 and 7.

(5) A method of forming a blade utilized in a turbine engine assembly, the method, according to another exemplary embodiment of this disclosure, among other possible things includes forming an airfoil of the blade assembly to include a thickness between pressure side and a suction side that is perpendicular to a camber line that includes a suction side thickness between the camber line and the suction side and a pressure side thickness between the camber line and the pressure side such that a maximum ratio of the pressure side thickness to the suction side thickness is between 3 and 7.

(6) Although the different examples have the specific components shown in the illustrations, embodiments of this disclosure are not limited to those particular combinations. It is possible to use some of the components or features from one of the examples in combination with features or components from another one of the examples.

(7) These and other features disclosed herein can be best understood from the following specification and drawings, the following of which is a brief description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic view of an example gas turbine engine.

(2) FIG. **2** is a schematic view of an example rotor embodiment.

(3) FIG. **3** is a schematic view of an example blade embodiment.

(4) FIG. **4** is a schematic cross-sectional view of an example airfoil embodiment.

(5) FIG. **5** is a schematic view of an example airfoil embodiment.
(6) FIG. **6** is a graph illustrating an example thickness ratio profile of an example airfoil embodiment.

DETAILED DESCRIPTION

(7) FIG. **1** schematically illustrates a gas turbine engine **20**. The disclosed gas turbine engine **20** includes airfoils with a defined thickness profile that provides a reduction in a pressure difference across an airfoil tip between pressure and suction sides of the airfoil. The airfoil thickness profile is asymmetric about a line of curvature **56** (FIG. **4**) that minimizes the pressure difference across the tip. The line of curvature defines a curvature of an airfoil, similar to a camber line, but with differing thickness on each of the pressure side and suction side. Reduction in the pressure difference across the airfoil tip between the pressure and suction sides of the airfoil provides a reduction in leakage flows.
(8) The example gas turbine engine **20** is a turbofan that generally incorporates a fan section **22**, a compressor section **24**, a combustor section **26** and a turbine section **28**. The fan section **22** drives air along a bypass flow path B in a bypass duct defined within a nacelle **30**. The compressor section **24** drives air along a core flow path C into the compressor section **24** for compression and communication into the combustor section **26**. In the combustor section **26**, the compressed air is mixed with fuel from a fuel system **32** and burnt to generate an exhaust gas flow that expands through the turbine section **28** to generate mechanical power utilized to drive the fan section **22** and the compressor section **26**.
(9) Although depicted as a turbofan turbine engine in the disclosed non-limiting embodiment, it should be understood that the concepts described herein are not limited to use with turbofans as the teachings may be applied to other types of gas turbine engines.
(10) The compressor section **24** includes at least one compressor rotor **34** and the turbine section **26** includes at least one turbine rotor **38**. The rotors **34**, **38** includes blades that rotate relative to a fixed case structure. Leakage flows between tips of the blades reduces engine efficiency. Some clearance is required to accommodate relative movement during operation and therefore limits reduction in physical clearance between blade tips and a fixed case structure. Example airfoil embodiments according to this disclosure include a defined thickness profile that reduces pressure differentials across tips of the blades that reduce leakage flows while enabling minimum clearances.
(11) Referring to FIGS. **2** and **3** with continued reference to FIG. **1**, the rotors **34**, **38** are shown schematically and include a plurality of blades **36** supported on a hub structure **40**. The example blades **36** may be compressor blades or turbine blades and are shown schematically by way of example. The blades **36** rotate relative to a fixed case structure that is shown schematically at **70**. A leakage flow **74** moves through a clearance gap **72** between a tip of the blade **36** and the case structure **70**.
(12) Each of the blades **36** include an airfoil **42** with a pressure side **52** and a suction side **54**. Each side of the airfoil **42** extends between a leading edge **44**, a trailing edge **46**, a base **50** and a tip **48**.
(13) Referring to FIG. **4** with continued reference to FIGS. **2** and **3**, a cross-section of an example airfoil **36** is shown and illustrates a line of curvature **56** that extends from the leading edge **44** to the trailing edge **46**. The airfoil **42** has a thickness **58** that is measured perpendicular to the line of curvature **56**. The thickness **58** between the pressure side **52** and the suction side **54** varies along the line of curvature **56** between the leading edge **44** and the trailing edge **46**.
(14) The thickness **58** is a total thickness across the airfoil **42** and is formed from a pressure side thickness **60** and a suction side thickness **62**. The pressure side thickness **60** is disposed perpendicular to line of curvature **56** between the line of curvature **56** and the pressure side **52**. The suction side thickness **62** is disposed perpendicular to the line of curvature **56** between the line of curvature **56** and the suction side **54**.
(15) The example airfoil **42** includes a thickness profile along the line of curvature **56** where the pressure side thickness **60** is greater than the suction side thickness **62**. The larger pressure side

thickness **60** provides for a bias of the airfoil thickness **58** toward the pressure side **52**. The biased thickness toward the pressure side **52** generates a bump that provides localized increases in flow velocity that reduce pressure on the pressure side **52**. Reducing pressure on the pressure side **52** provides a reduction in the pressure differential between the pressure side **52** and the suction side **54**. The reduced pressure difference provides a reduction in leakage flow **74** without changing the clearance gap **72**.
(16) Referring to FIGS. **5** and **6** with continued reference to FIG. **4**, an example airfoil **42** embodiment includes a thickness profile **78** defined as a thickness ratio **80** between the pressure side thickness **60** and the suction side thickness **62** as shown in the graph **76** of FIG. **6**.
(17) In one example embodiment, the example thickness profile **78** is disposed along the airfoil within a tip region **82** as schematically shown in FIG. **5**. In one example embodiment, the tip region **82** is disposed between 80% and 100% of the airfoil height **64**. Although the example thickness profile **78** is disclosed as being located within the tip region **82**, the thickness profile **78** may be located throughout the airfoil height **64** and remain within the contemplation of this disclosure.
(18) The thickness profile **78** illustrates variations in the ratio of the pressure side thickness **60** to the suction side thickness **62** for locations along the meridional length **66**. In one example embodiment, the thickness ratio **80** of the pressure side thickness **62** to the suction side thickness is greatest within the meridional length **66** between 5% and 40%. In another example embodiment, the thickness ratio **80** is greatest within a meridional length between 10% and 30%. In another disclosed example embodiment, the thickness ratio **80** is greatest between 15% and 25% of the meridional length **66**. In another disclosed example embodiment, the thickness ratio **80** is greatest at 20% of the meridional length **66**.
(19) The thickness ratio **80** reflects the increase in the pressure side thickness **60** compared to the suction side thickness **62**. The example thickness profile **78** is shown with upper and lower limits **86**, **90** centered along a mean thickness ratio **92**. In one example embodiment, the thickness ratio **80** varies between 3 and 7 along the meridional length **66**. In another example embodiment, a maximum thickness ratio **84** is between 5 and 7. In other words, the pressure side thickness **60** is between 5 and 7 times larger than the suction side thickness **62**. In another example embodiment, the maximum thickness ratio **84** is 6.
(20) The example thickness ratio **80** may vary along the meridional length **66** but maintains the overall thickness **58**. Maintaining the overall thickness **58** does not change the physical properties of the airfoil **42** and thereby maintains the defined mechanical structural properties. Because the airfoils thickness **58** remains unchanged, airfoils structural integrity is not compromised nor changed. Moreover, maintaining the same overall thickness **58** of the airfoil **42** has minimal impact on cost and/or manufacture.
(21) A turbine engine assembly **20** according to an exemplary embodiment of this disclosure, among other possible things includes at least one rotor that has a plurality of blades **36**, each of the blades **36** includes an airfoil **42** that has a pressure side **52** and a suction side **54** that each extend between a leading edge **44**, a trailing edge **46**, a tip **48** and a base **50**. The airfoil **42** has a thickness between the pressure side **52** and the suction side **54** perpendicular to a line of curvature **56** that varies between the leading edge **44** and the trailing edge **46**. The thickness includes a suction side thickness **62** between the line of curvature **56** and the suction side **54** and a pressure side thickness **60** between the line of curvature **56** and the pressure side **52**. A maximum ratio of the pressure side thickness **60** to the suction side thickness **62** is between 3 and 7.
(22) In a further embodiment of the foregoing turbine engine assembly, the maximum ratio of the pressure side thickness **60** to the suction side thickness **62** is disposed between 80% and 100% of a height of the airfoil **42** between the base **50** and the tip portion.
(23) In a further embodiment of any of the foregoing turbine engine assemblies, the maximum ratio of the pressure side thickness **60** to the suction side thickness **62** is at a location between 5% and

40% of a meridional length **66** between the leading edge **44** and the trailing edge **46**.
(24) In a further embodiment of any of the foregoing turbine engine assemblies, the maximum ratio of the pressure side thickness **60** to the suction side thickness **62** is at a location between 10% and 30% of a meridional length **66** between the leading edge **44** and the trailing edge **46**.
(25) In a further embodiment of any of the foregoing turbine engine assemblies, the maximum ratio of the pressure side thickness **60** to the suction side thickness **62** is between 5 and 7.
(26) In a further embodiment of any of the foregoing turbine engine assemblies, the maximum ratio of the pressure side thickness **60** to the suction side thickness **62** is 6.
(27) In a further embodiment of any of the foregoing turbine engine assemblies, the maximum ratio of the pressure side thickness **60** to the suction side thickness **62** is at a location that is between 15% and 25% of the meridional length **66** between the leading edge **44** and the trailing edge **46**.
(28) In a further embodiment of any of the foregoing turbine engine assemblies, the at least one rotor includes a compressor rotor **34**.
(29) In a further embodiment of any of the foregoing turbine engine assemblies, the at least one rotor includes a turbine rotor **38**.
(30) A blade **36** for compressor section of a turbine engine assembly according to another exemplary embodiment of this disclosure, among other possible things includes an airfoil **42** that has a pressure side **52** and a suction side **54** that each extend between a leading edge **44**, a trailing edge **46**, a tip **48** and a base **50**. The airfoil **42** has a thickness between the pressure side **52** and the suction side **54** perpendicular to a line of curvature **56** that varies between the leading edge **44** and the trailing edge **46**. The thickness includes a suction side thickness **62** between the line of curvature **56** and the suction side **54** and a pressure side thickness **60** between the line of curvature **56** and the pressure side **52**. A maximum ratio of the pressure side thickness **60** to the suction side thickness **62** is between 3 and 7.
(31) In a further embodiment of the foregoing blade, the ratio of the pressure side thickness **60** to the suction side thickness **62** is disposed between 80% and 100% of a height of the airfoil **42** between the base **50** and the tip portion.
(32) In a further embodiment of any of the foregoing blades, the maximum ratio of the pressure side thickness **60** to the suction side thickness **62** is at a location between 5% and 40% of a meridional length **66** between the leading edge **44** and the trailing edge **46**.
(33) In a further embodiment of any of the foregoing blades, the maximum ratio of the pressure side thickness **60** to the suction side thickness **62** is at a location between 10% and 30% of a meridional length **66** between the leading edge **44** and the trailing edge **46**.
(34) In a further embodiment of any of the foregoing blades, the maximum ratio of the pressure side thickness **60** to the suction side thickness **62** is between 5 and 7.
(35) In a further embodiment of any of the foregoing blades, the maximum ratio of the pressure side thickness **60** to the suction side thickness **62** is 6.
(36) In a further embodiment of any of the foregoing blades, the maximum ratio of the pressure side thickness **60** to the suction side thickness **62** is at a location that is between 15% and 25% of the meridional length **66** between the leading edge **44** and the trailing edge **46**.
(37) A method of forming a blade utilized in a turbine engine assembly, the method, according to another exemplary embodiment of this disclosure, among other possible things includes forming an airfoil **42** of the blade assembly to include a thickness between pressure side **52** and a suction side **54** that is perpendicular to a line of curvature **56** that includes a suction side thickness **62** between the line of curvature **56** and the suction side **54** and a pressure side thickness **60** between the line of curvature **56** and the pressure side **52** such that a maximum ratio of the pressure side thickness **60** to the suction side thickness **62** is between 3 and 7.
(38) In a further embodiment of the foregoing, the method further includes locating the maximum ratio of the pressure side thickness **60** to the suction side **54** thickness within a location between 80% and 100% of a height of the airfoil **42** between a base **50** and a tip portion.

(39) In a further embodiment of any of the foregoing, the method further includes locating the maximum ratio of the pressure side thickness **60** to the suction side thickness **62** within a location between 5% and 40% of a meridional length **66** between a leading edge **44** and a trailing edge **46**.
(40) In a further embodiment of any of the foregoing methods, the maximum ratio of the pressure side thickness **60** to the suction side thickness **62** is between 5 and 7 and is within a location between 15% and 25% of the meridional length **66** between a leading edge **44** and a trailing edge **46**.
(41) Although an example embodiment has been disclosed, a worker of ordinary skill in this art would recognize that certain modifications would come within the scope of this disclosure. For that reason, the following claims should be studied to determine the scope and content of this disclosure.

Claims

1. A turbine engine assembly comprising: at least one rotor having a plurality of blades, wherein each blade includes; an airfoil having; a pressure side and a suction side that each extend between a leading edge, a trailing edge, a tip, and a base; and a thickness between the pressure side and the suction side perpendicular to a line of curvature that defines a curvature of the airfoil, wherein the thickness varies between the leading edge and the trailing edge, wherein the thickness comprises a suction side thickness between the line of curvature and the suction side and a pressure side thickness between the line of curvature and the pressure side, wherein a ratio of the pressure side thickness to the suction side thickness continually increases from the leading edge toward a maximum ratio of the pressure side thickness to the suction side thickness and the maximum ratio of the pressure side thickness to the suction side thickness is between 3 and 7.
2. The turbine engine assembly as recited in claim 1, wherein the maximum ratio of the pressure side thickness to the suction side thickness is disposed between 80% and 100% of a height of the airfoil between the base and the tip.
3. The turbine engine assembly as recited in claim 2, wherein the maximum ratio of the pressure side thickness to the suction side thickness is at a location between 5% and 40% of a meridional length between the leading edge and the trailing edge.
4. The turbine engine assembly as recited in claim 2, wherein the maximum ratio of the pressure side thickness to the suction side thickness is at a location between 10% and 30% of a meridional length between the leading edge and the trailing edge.
5. The turbine engine assembly as recited in claim 1, wherein the maximum ratio of the pressure side thickness to the suction side thickness is between 5 and 7.
6. The turbine engine assembly as recited in claim 3, wherein the maximum ratio of the pressure side thickness to the suction side thickness is 6.
7. The turbine engine assembly as recited in claim 6, wherein the maximum ratio of the pressure side thickness to the suction side thickness is at a location that is between 15% and 25% of the meridional length between the leading edge and the trailing edge.
8. The turbine engine assembly as recited in claim 1, wherein the at least one rotor comprises a compressor rotor.
9. The turbine engine assembly as recited in claim 1, wherein the at least one rotor comprises a turbine rotor.
10. A blade for compressor section of a turbine engine assembly comprising: an airfoil having a pressure side and a suction side that each extend between a leading edge, a trailing edge, a tip and a base, the airfoil having a thickness between the pressure side and the suction side perpendicular to a line of curvature that defines a curvature of the airfoil, wherein the thickness that varies between the leading edge and the trailing edge, wherein the thickness comprises a suction side thickness between the line of curvature and the suction side and a pressure side thickness between the line of

curvature and the pressure side, wherein a ratio of the pressure side thickness to the suction side thickness continually increases from the leading edge toward a maximum ratio of the pressure side thickness to the suction side thickness and the maximum ratio of the pressure side thickness to the suction side thickness is between 3 and 7.
11. The blade as recited in claim 10, wherein the ratio of the pressure side thickness to the suction side thickness is disposed between 80% and 100% of a height of the airfoil between the base and the tip.
12. The blade as recited in claim 11, wherein the maximum ratio of the pressure side thickness to the suction side thickness is at a location between 5% and 40% of a meridional length between the leading edge and the trailing edge.
13. The blade as recited in claim 11, wherein the maximum ratio of the pressure side thickness to the suction side thickness is at a location between 10% and 30% of a meridional length between the leading edge and the trailing edge.
14. The blade as recited in claim 10, wherein the maximum ratio of the pressure side thickness to the suction side thickness is between 5 and 7.
15. The blade as recited in claim 10, wherein the maximum ratio of the pressure side thickness to the suction side thickness is 6.
16. The blade as recited in claim 15, wherein the maximum ratio of the pressure side thickness to the suction side thickness is at a location that is between 15% and 25% of the meridional length between the leading edge and the trailing edge.
17. A method of forming a blade utilized in a turbine engine assembly, the method comprising: forming an airfoil of the blade to include a thickness between a pressure side and a suction side that is perpendicular to a line of curvature that defines a curvature of the airfoil, wherein the thickness includes a suction side thickness between the line of curvature and the suction side and a pressure side thickness between the line of curvature and the pressure side such that a ratio of the pressure side thickness to the suction side thickness continually increases from the leading edge toward a maximum ratio of the pressure side thickness to the suction side thickness and the maximum ratio of the pressure side thickness to the suction side thickness is between 3 and 7.
18. The method as recited in claim 17, further comprising locating the maximum ratio of the pressure side thickness to the suction side thickness within a location between 80% and 100% of a height of the airfoil between a base and a tip portion.
19. The method as recited in claim 18, further comprising locating the maximum ratio of the pressure side thickness to the suction side thickness within a location between 5% and 40% of a meridional length between a leading edge and a trailing edge.
20. The method as recited in claim 18, wherein the maximum ratio of the pressure side thickness to the suction side thickness is between 5 and 7 and is within a location between 15% and 25% of the meridional length between a leading edge and a trailing edge.